

Institute/Department	School of Computer Science Engineering	Program	Bachelor of Technology (Computer Science and Engineering-Artificial Intelligence and Machine Learning) with Microsoft(LCS202)
Master Subject Coordinator Name:	Vinayak	Master Subject Coordinator E-Code:	L100943
Course Name	Software Engineering	Course Code	25CST-217

Lecture	Tutorial	Practical	Self Study	Credit	Subject Type
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Course Type	Course Category	Mode of Assessment	Mode of Delivery
Major Core	Graded (GR)	Theory Examination (ET)	Theory (TH)

Mission of the Department	M1. Develop and deliver comprehensive curricula that equip students with the knowledge and skills necessary for successful careers in computer science and engineering areas. M2. Implement innovative teaching methodologies to foster critical thinking and problem-solving abilities among students. M3. Encourage faculty and students to engage in pioneering research projects that contribute to technological advancements. M4. Facilitate partnerships with other departments and institutions to conduct convergent research that bridges multiple disciplines. M5. Organize seminars, workshops, and collaborative projects that encourage cross-disciplinary engagement. M6. Establish platforms for alumni to actively participate in departmental activities, such as mentoring programs and guest lectures. M7. Maintain regular communication with alumni to foster a supportive and engaged community.
Vision of the Department	V1. Provides undergraduate and graduate students with a high-quality education leading to rewarding, high-paying careers. V2. Engages faculty in cutting-edge research and innovation, actively involving and training students in these endeavors. V3. Serves as a hub where convergent research themes intersect, enabling advancements across various disciplines. V4. Maintains active engagement with alumni, fostering a strong and supportive community

Program Educational Objectives(PEOs)

PEO1	Graduates of Computer Science and Engineering will contribute to the Nation's growth through their ability to solve diverse and complex computer science and engineering problems across a broad range of application areas.
PEO2	Graduates of Computer Science and Engineering will be successful professionals, designing and implementing Products & Services of global standards in the field of Computer Science & Engineering, becoming entrepreneurs, Pursuing higher studies & research.
PEO3	Graduates of Computer Science and Engineering Program will be able to adapt to changing scenarios of dynamic technology with an ability to solve larger societal problems using logical and flexible approach in decision making.

Program Specific OutComes(PSOs)

PSO1	Apply core concepts of computer science, including programming, algorithms, database systems, and software engineering, to design and develop efficient and innovative computational solutions.
PSO2	Integrate Artificial Intelligence and Machine Learning techniques into software and system designs to enhance functionality, enable intelligent automation, and solve complex real- world challenges

Program OutComes(POs)

PO1	Ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problem
PO2	Ability to identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	Ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.

PO4	Ability to conduct investigation into complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	Ability to create, select and apply appropriate techniques, resources and modern engineering, and IT tools, including prediction and modelling, to complex engineering activities, with an understanding of the limitations.
PO6	Ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	Ability to recognize and incorporate the diversity and commonalities of engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
PO8	Ability to apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PO9	Ability to function effectively as an individual and as member or leader in diverse teams and interdisciplinary settings.
PO10	Ability to communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Ability to demonstrate knowledge and understanding of the engineering and management principles and also apply these principles to one's own work as member and leader to the teams to manage projects and interdisciplinary teams.
PO12	Ability to recognize the need for, and have the preparations and ability to engage in independent and lifelong learning in the broadest context of technological change.

Text Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	An Integrated Approach to Software Engineering	Pankaj Jalote	THIRD	Narosa Pub	2005
2	Fundamentals of Software Engineering	Rajib Mall	FIFTH	PHI Learning	2015
3	Software Engineering: A Practitioner's Approach	R S. Pressman	NINE	McGraw-Hill	2006
4	Software Engineering	Sommerville	9th	Pearson Education	2019
5	Software Engineering & Project Managements	Richard H.Thayer	One	WileyIndia	2018
6	Software Engineering	Waman S.Jawadekar	FIRST	TMH	2020

Reference Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Software Project Management	Bob Hughes, M.Cotterell, Rajib Mall	FIFTH	McGrawHill	2018

Course OutCome	
SrNo	OutCome
CO1	Compare software development models and framework activities for software development
CO2	Explain Software Requirement Specification with functional and non-functional requirements.
CO3	Analyse the software in the graphical form of DFD, sequence diagram, ERD, activity diagram using suitable example.
CO4	Explain various software testing techniques.
CO5	Estimate cost and efforts required for design and development of software project.

Lecture Plan Preview-Theory

Unit No	LectureNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Numer (s)
1	1	Introduction to Software Product and Software Process	Software Product, Types of Software Products, Software Process,Product vs Process,Software Product Characteristics	T-Fundamentals of Software Engin	PPT	CO1
1	2	Classical Software Process Models	Linear Sequential (Waterfall) Model, Advantages and Limitations, Applications of Waterfall Model, Prototype Model	T-Fundamentals of Software Engin	PPT	CO1
1	3	Advanced Software Process Models	RAD (Rapid Application Development) Model,Comparison of Waterfall, Prototype, and RAD Models,Model Selection Criteria	T-Software Engineering: A Practi	PPT,Video Lecture	CO1
1	4	Evolutionary Process Models	Incremental Model, Spiral Model,Risk Analysis in the Spiral Model,Comparison of Evolutionary Models	T-Software Engineering	PPT,Video Lecture	CO1
1	5	Modern Software Development Approaches	Component Assembly Model, Rational Unified Process (RUP), Agile Process Models,Agile Principles and Practices	T-Fundamentals of Software Engin	PPT,Video Lecture	CO1
1	6	Software Process Improvement and Metrics	Software Process Customization,Software Process Improvement,Capability Maturity Model (CMM),Product Metrics,Process Metrics,Case Studies and Revision	T-Software Engineering & Project	Case Study,PPT,V ideo Lecture	CO1
2	7	Introduction to Software Requirements	Definition of Requirements, Importance of Requirements Engineering, Types of Requirements, Functional Requirements, Non-Functional Requirements,Characteristics of Good Requirements	T-Software Engineering	PPT,Video Lecture	CO2
2	8	Requirement Sources and Elicitation Techniques	Requirement Sources, Existing Systems, Regulatory Bodies,Requirement Elicitation Process,Elicitation Techniques	T-Fundamentals of Software Engin	PPT,Video Lecture	CO2
2	9	Requirement Analysis and Modeling	Requirement Analysis Process Feasibility Analysis Requirement Prioritization Requirement Negotiation Analysis Principles Introduction to Analysis Models	T-Software Engineering	PPT,Video Lecture	CO2
2	10	Function-Oriented Analysis Modeling	Structured Analysis Approach Context Diagram Data Flow Diagrams (DFD) Data Dictionary Process Specifications (PSPEC) Advantages and Limitations of Function-Oriented Modeling	T-Software Engineering	PPT,Video Lecture	CO2
2	11	Object-Oriented Analysis and Use Case Modeling	Object-Oriented Analysis Concepts Classes, Objects, Attributes, and Methods UML Overview Use Case Modeling,Use Case Diagrams and Descriptions	,T-Software Engineering ,R-Software Project Management	Case Study,PPT,V ideo Lecture	CO2

2	12	Software Requirement Specification and Validation	System Requirement Specification (SyRS) Software Requirement Specification (SRS) Structure and Contents of SRS Document IEEE SRS Standards Requirement Validation Techniques, Requirement Management and Traceability	,T-Software Engineering: A Practi,R-Software Project Management	PPT,Video Lecture	CO2
3	13	Introduction to Software Design Process and Design Principles	Software Design Process Importance of Software Design Design Concepts,Design Principles and Guidelines Characteristics of Good Software Design	,T-Software Engineering: A Practi,R-Software Project Management	PPT,Video Lecture	CO3
3	14	Software Modeling and UML	Software Modeling Concepts Unified Modeling Language (UML) Overview UML Diagrams,Role of UML in Software Design	,T-Software Engineering ,R-Software Project Management	PPT,Video Lecture	CO3
3	15	Architectural Design	Software Architecture Fundamentals Architectural Design Process Architectural Views Architectural Styles	,T-Software Engineering ,T-Software Engineering & Project,R-Software Project Management	Case Study,PPT,V ideo Lecture	CO3
3	16	User Interface Design and Function-Oriented Design	User Interface (UI) Design Principles User-Centered Design Interface Design Process Function-Oriented Design Structured Analysis and Structured Design (SA/SD) Structure Charts Design Documentation	T-Software Engineering & Project,T-Software Engineering: A Practi	PPT,Video Lecture	CO3
3	17	Component-Based Design and Design Metrics	Component-Based Software Design (CBSD) Reusable Components Component Interfaces and Integration Design Quality Attributes Design Metrics,Evaluation of Software Design Quality	T-Software Engineering ,T-Software Engineering: A Practi	PPT,Video Lecture	CO3
4	18	Software Analysis and Code Inspection	Software Analysis Overview Static Analysis,Dynamic Analysis,Code Inspections and Walkthroughs Benefits of Early Defect Detection	,T-Software Engineering & Project,T-Software Engineering: A Practi,R-Software Project Management	Infographics, PPT,Video Lecture	CO4
4	19	Fundamentals of Software Testing	Introduction to Software Testing Objectives and Principles of Testing Verification vs Validation Software Test Process Testing Life Cycle (STLC) Testing Levels Overview Role of Testing in Software Quality Assurance	T-Fundamentals of Software Engin,T-Software Engineering	PPT,Video Lecture	CO4
4	20	Test Design and Test Techniques	Test Criteria Test Case Design Test Oracles Test Data Preparation Black-Box Testing Techniques,Introduction to White-Box Testing	T-Software Engineering	PPT,Video Lecture	CO4
4	21	Unit Testing and Integration Testing	White-Box Unit Testing Control Flow Testing Statement, Branch, and Path Coverage Unit Testing Frameworks,Integration Testing,Stub and Driver Concepts	,T-Software Engineering ,T-Software Engineering: A Practi,R-Software Project Management	PPT,Simulati on,Video Lecture	CO4

4	22	System Testing and Test Management	System Testing Acceptance Testing Regression Testing Performance Testing Security Testing Reliability Testing Test Plan Preparation Test Metrics and Measurement Software Testing Tools	,T-Software Engineering ,T-Software Engineering &; Project,T-Software Engineering: A Practi,R-Software Project Management	Case Study,PPT,V ideo Lecture	CO4
4	23	Introduction to Object-Oriented Analysis and Design	Fundamentals of Object-Oriented Analysis Object-Oriented Design Concepts,UML in OOAD Comparison of Object-Oriented and Structured Software Development Advantages and Applications of OOAD	T-An Integrated Approach to Soft,T-Software Engineering ,T-Software Engineering &; Project,T-Software Engineering: A Practi	Instructor Lead WorkShop,P PT,Video Lecture	CO4
5	24	Software Maintenance Fundamentals	Need for Software Maintenance Software Evolution Types of Maintenance,Maintenance Process and Challenges	,T-Fundamentals of Software Engin,T-Software Engineering ,T-Software Engineering: A Practi,R-Software Project Management	Case Study,PPT,V ideo Lecture	CO5
5	25	Software Configuration and Change Management	Software Configuration Management (SCM) SCM Activities,Software Change Management Change Request Process Change Control Board (CCB)	,T-Fundamentals of Software Engin,T-Software Engineering &; Project,R-Software Project Management	PPT,Video Lecture	CO5
5	26	Version Control and Program Comprehension	Version Control Concepts Version Control Systems,Change Control Process Program Comprehension Techniques Software Documentation and Maintenance Support	,T-Fundamentals of Software Engin,T-Software Engineering ,R-Software Project Management	PPT,Video Lecture	CO5
5	27	Re-engineering and Reverse Engineering	Software Re-engineering Concepts Re-engineering Process Benefits and Challenges of Re-engineering Reverse Engineering Difference between Re-engineering and Reverse Engineering Tool Support for Maintenance and Re-engineering	,T-Software Engineering &; Project,T-Software Engineering: A Practi,R-Software Project Management	PPT,Video Lecture	CO5
5	28	Software Project Management and Feasibility Analysis	Introduction to Software Project Management Project Management Concepts Feasibility Analysis,Project Life Cycle	,T-An Integrated Approach to Soft,T-Software Engineering ,T-Software Engineering &; Project,R-Software Project Management	PPT,Video Lecture	CO5
5	29	Project Planning, Estimation, and Scheduling	Project and Process Planning Resource Allocation and Management Software Effort Estimation Cost Estimation Techniques Schedule Estimation Project Scheduling Methods,Project Tracking and Monitoring	,T-Fundamentals of Software Engin,T-Software Engineering ,T-Software Engineering &; Project,R-Software Project Management	Activity,PPT, Video Lecture	CO5
5	30	Risk Management, Software Quality Assurance, and Metrics	Risk Assessment and Risk Mitigation Risk Management Process Software Quality Assurance (SQA) Quality Standards and Reviews Software Quality Control Project Plan Documentation Project Metrics,Measurement and Reporting	,T-Software Engineering ,T-Software Engineering &; Project,T-Software Engineering: A Practi,R-Software Project Management	PPT,Video Lecture	CO5

Sr No	Assessment Name	Exam Name	Max Marks

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	NA	2	NA	NA	NA	2	2	3	NA	3	1
CO2	3	3	3	1	1	NA	NA	NA	2	2	3	NA	2	1
CO3	3	3	3	1	1	NA	NA	NA	3	2	3	NA	2	1
CO4	3	3	3	NA	1	NA	NA	NA	3	2	3	NA	2	1
CO5	3	3	3	NA	1	NA	NA	NA	3	NA	3	NA	2	1
Target	3	3	3	1	1.2	NA	NA	NA	2.6	2	3	NA	2.2	1

